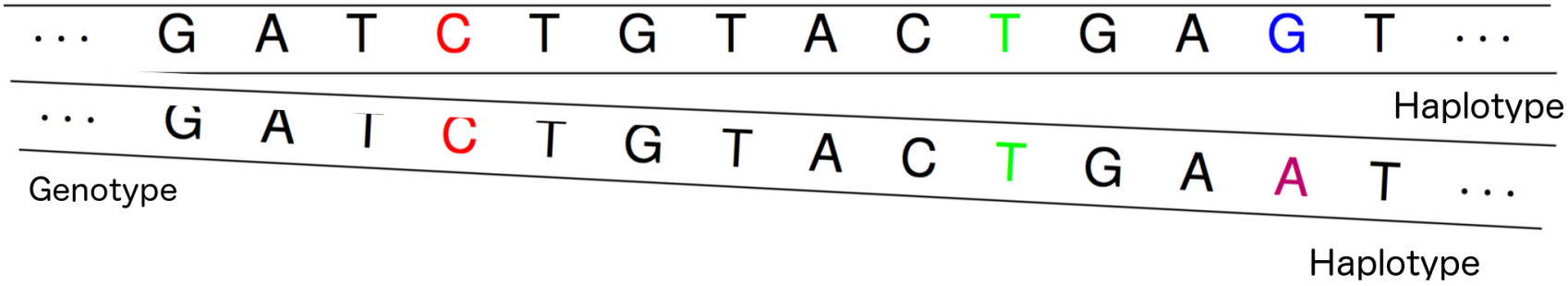
Haplotype Inference

Haplotype Inference

- Genes are packaged in bundles called chromosomes. (Chromosomes are therefore partitions of DNA)
- In Diploid organisms (e.g., animals) there are almost identical chromosome pairs. Each pair is made of an inherited chromosome from the father and another one from the mother.
- A haplotype is a DNA sequence that has been inherited from one parent.
- A genotype is made of a pair of two corresponding haplotypes.



Haplotype Inference

Each individual inherits two haplotypes (from the mother and from the father) and for most regions of the genome they are identical.

···	G	A	T	C	T	G	T	A	C	T	G	A	G	T	···
···	G	A	T	C	T	G	T	A	C	T	G	A	A	T	···
				⇑						⇑			⇑		

In some typical positions, the bases are subject to mutations.

In the most common case, there is a Single Nucleotide Polymorphism (SNP).

Possible mutations are $C \leftrightarrow T$ and $A \leftrightarrow G$

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Haplotype Inference

- The genome (genotype) is easier and cheaper to be obtained.
- We need computational methods to guess haplotypes
- The Haplotype Inference problem is introduced to investigate genetic variations in a population.
- It informs us about heritability and some parent's genetic features
- Typically SNPs sites are the target of the analysis

Each individual carries two haplotypes (from the mother and from the father)

G	A	A	T	C	T	T	C	G	T	A	C	T	G	A	G	T
G	A	A	T	C	T	T	C	G	T	A	C	T	G	A	A	T

Let us focus on the SNPs:

A	C	T	G
A	C	T	A

We encode SNPs according to: $A \mapsto 0$ $C \mapsto 0$ $G \mapsto 1$ $T \mapsto 1$

0	0	1	1
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0	0	1	2

The genotype is set to 2 if there is a mismatch

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Goal

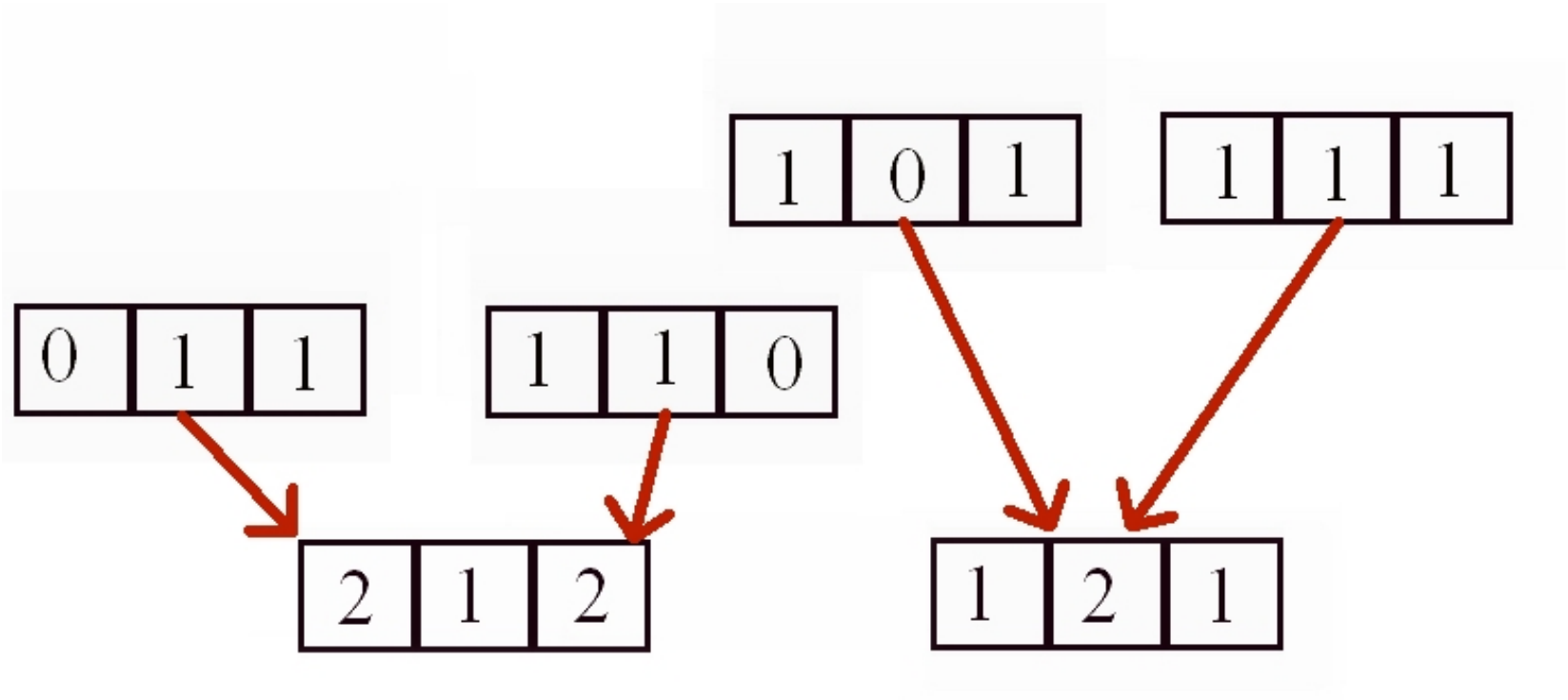
- Assume we are able to collect a set of genotypes
- We would like to guess the set of haplotypes in the population that can express all the genotypes
- The set should be minimal (there is a biological reason, see below)

Haplotype Inference

2	1	2

1	2	1

Haplotype Inference



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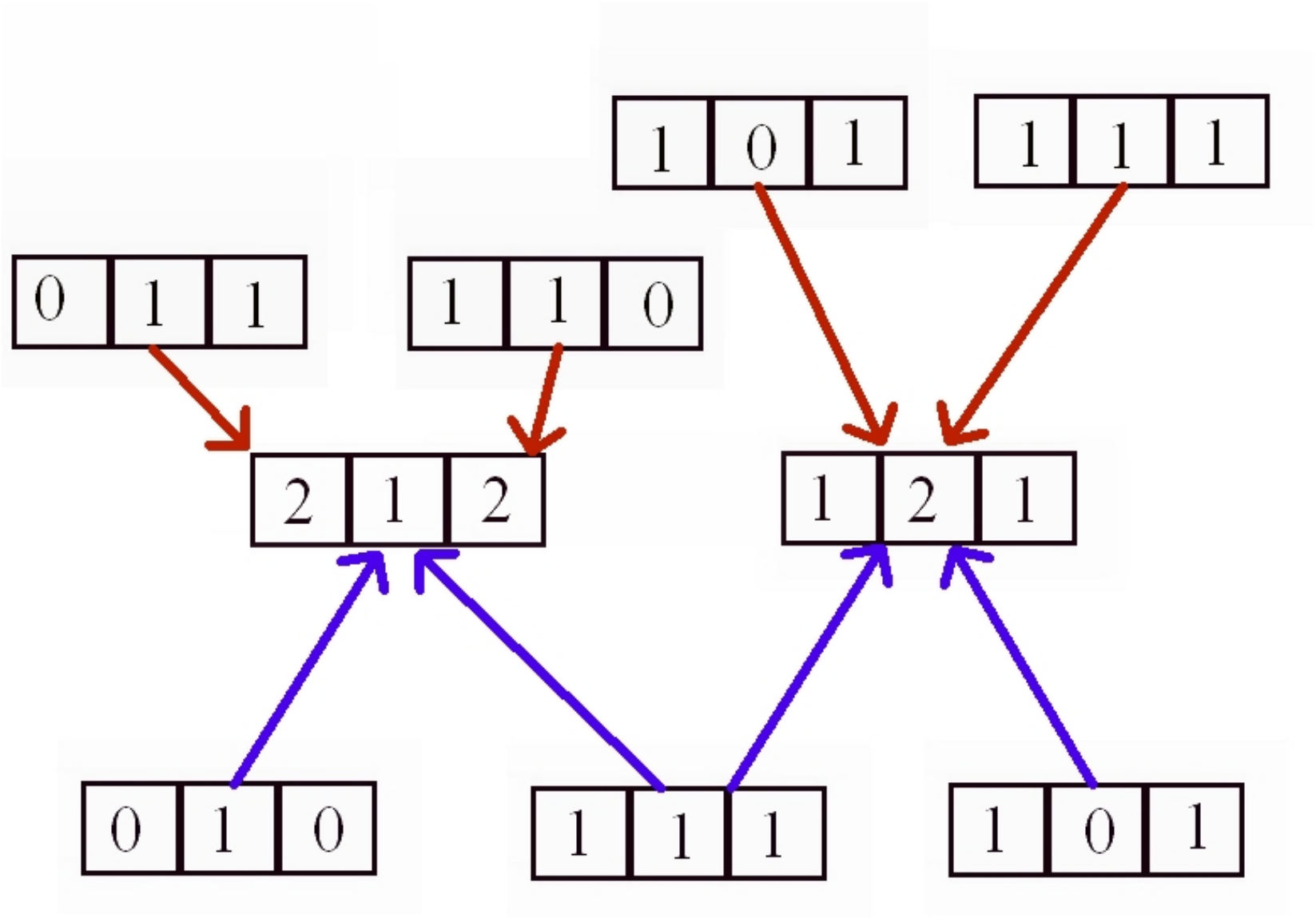

Haplotype Inference

- A string of $\{0,1\}^*$ is called a *haplotype*
- A string of $\{0,1,2\}^*$ is called a *genotype*
- Two equal length haplotypes generate a unique genotype
- The rules are $0 \oplus 0 = 0$, $1 \oplus 1 = 1$, $0 \oplus 1 = 1 \oplus 0 = 2$
  E.g., $0010, 0101 \Rightarrow 0222$
- If we have a genotype, we can only conjecture (potentially exponentially many) pairs of *haplotypes* that generated it (observe that, e.g., $0110, 0001 \Rightarrow 0222$)
- DNA sequencing allows us to know genotypes!
- Since genotypes are introduced in evolution, it is reasonable to find minimal sets of haplotypes explaining the known genotypes.

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Haplotype Inference

- Let $H \subseteq \{0,1\}^n$ be a set of *haplotypes* and
- $G \subseteq \{0,1,2\}^n$ be a set of *genotypes*.
- Given $h_1, h_2 \in H$ and $g \in G$, $\{h_1, h_2\}$ explains g if and only if $|h_1| = |h_2| = |g|$ and $\forall i \in [1..n]$:

$$\begin{array}{lcl} g[i] \leq 1 & \longrightarrow & h_1[i] = h_2[i] = g[i] \\ g[i] = 2 & \longrightarrow & h_1[i] \neq h_2[i] \end{array}$$

- A set of haplotypes H explains a set of genotypes G if for all $g \in G$ there are $h_1, h_2 \in H$ such that $\{h_1, h_2\}$ explains g.
- Given a set of genotypes G and an integer k, the *haplotype inference problem (HIP)* by pure parsimony is the problem of finding a set H that explains G and such that $|H| = k$ (decision version: NP complete).